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HYDROGEN FUEL SYSTEM, A HYDROGEN INTERNAL COMBUSTION ENGINE SYSTEM, AND A METHOD FOR OPERATING A HYDROGEN FUEL SYSTEM

Abstract

A hydrogen fuel system includes a hydrogen fuel conduit, a hydrogen fuel rail, and a vent valve. The hydrogen fuel rail is in fluid communication with the hydrogen fuel conduit. The vent valve is in fluid communication with at least one of the hydrogen fuel conduit or the hydrogen fuel rail. The vent valve is configured to selectively transition between an open state and a closed state, the open state facilitating at least one of venting or purging of the hydrogen fuel system.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application claims priority to and the benefit of Indian Provisional Patent Application No. 202441013666, filed Feb. 26, 2024, and Indian Nonprovisional patent Application No. 202441013666, filed Feb. 20, 2025. The contents of these applications are incorporated by reference in its entirety.

TECHNICAL FIELD

[0002] The present disclosure relates generally to hydrogen internal combustion engines.

BACKGROUND

[0003] Hydrogen internal combustion engines combust air and hydrogen to produce power. These internal combustion engines may include a fuel system that leaks hydrogen into an environment surrounding the hydrogen internal combustion engine.

SUMMARY

[0004] In some implementations, a hydrogen fuel system includes a hydrogen fuel conduit, a hydrogen fuel rail, and a vent valve. The hydrogen fuel rail is in fluid communication with the hydrogen fuel conduit. The vent valve is in fluid communication with at least one of the hydrogen fuel conduit or the hydrogen fuel rail. The vent valve is configured to selectively transition between an open state and a closed state, the open state facilitating at least one of venting or purging of the hydrogen fuel system.

[0005] In some implementations, a method for operating a hydrogen fuel system includes receiving a valve signal and responsive to receiving the valve signal, transmitting, to a component of the hydrogen fuel system, an opening signal that causes a vent valve of the hydrogen fuel system to transition from a closed state to an open state. The open state facilitates at least one of venting or purging of the hydrogen fuel system.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0006] The disclosure will become more fully understood from the following detailed description, taken in conjunction with the accompanying figures, wherein like reference numerals refer to like elements, in which:

[0007] FIG. 1 is a block schematic diagram of a hydrogen internal combustion engine system, according to one embodiment;

[0008] FIG. 2 is a block schematic diagram of a hydrogen fuel system for a hydrogen internal combustion engine system, according to one embodiment;

[0009] FIG. 3 is a partial cross-sectional view of a vent valve for a hydrogen fuel system, according to one embodiment;

[0010] FIG. 4 is a partial cross-sectional view of another vent valve for a hydrogen fuel system, according to one embodiment;

[0011] FIG. 5 is a block schematic diagram of a hydrogen fuel system for a hydrogen internal combustion engine system, according to one embodiment; and

[0012] FIG. 6 is a block diagram of a method of operating a vent valve, according to one embodiment.

[0013] The foregoing and other features of the present disclosure will become apparent from the

following description and appended claims, taken in conjunction with the accompanying drawings. Understanding that these drawings depict only several embodiments in accordance with the disclosure and are therefore, not to be considered limiting of its scope, the disclosure will be described with additional specificity and detail through use of the accompanying drawings.

DETAILED DESCRIPTION

[0014] FIG. 1 depicts a hydrogen internal combustion engine system **100**. The hydrogen internal combustion engine system **100** may be implemented in a vehicle (e.g., car, truck, construction vehicle, municipal vehicle, military vehicle, maritime vessel, etc.), generator (e.g., genset, etc.), or other similar engine-based application.

[0015] The hydrogen internal combustion engine system **100** includes a hydrogen fuel system **104**. The hydrogen fuel system **104** includes a hydrogen fuel conduit **108**, a hydrogen fuel rail **112**, and a vent valve **118**. The hydrogen fuel rail **112** is in fluid communication with the hydrogen fuel conduit **108**. The vent valve **118** is in fluid communication with at least one of the hydrogen fuel conduit **108** or the hydrogen fuel rail **112**. The vent valve **118** is configured to selectively transition between an open state and a closed state, the open state facilitating at least one of venting or purging of the hydrogen fuel system **104**. The hydrogen internal combustion engine system **100** includes the hydrogen fuel system **104** and a hydrogen internal combustion engine **102**. The hydrogen internal combustion engine **102** combusts hydrogen and fluid to produce power. The hydrogen fuel rail **112** is coupled to the hydrogen internal combustion engine **102**.

[0016] In various embodiments, the hydrogen fuel system **104** includes the hydrogen fuel conduit **108**, the hydrogen fuel rail **112** in fluid communication with the hydrogen fuel conduit **108**, and the vent valve **118** in fluid communication with at least one of the hydrogen fuel conduit **108** or the hydrogen fuel rail **112**. The vent valve **118** is configured to selectively transition between an open state and a closed state. The open state facilitates at least one of venting or purging of the hydrogen fuel system **104**.

[0017] In various embodiments, a method for operating the hydrogen fuel system **104** includes receiving a valve signal and responsive to receiving the valve signal, transmitting, to a component of the hydrogen fuel system **104**, an opening signal that causes the vent valve **118** of the hydrogen fuel system **104** to transition from the closed state to the open state. The open state facilitates at least one of venting or purging of the hydrogen fuel system **104**.

[0018] In various embodiments, the hydrogen fuel system **104** also includes a compressed air pump **120** that is fluidly coupled to the vent valve **118** to facilitate operation of the vent valve **118**. Specifically, the compressed air pump **120** is configured to selectively provide compressed air to the vent valve **118** to transition the vent valve **118** between the open state and the closed state. In various embodiments, the hydrogen fuel system **104** also includes a controller **122**. The controller **122** is configured to cause the vent valve **118** to transition between the open state and the closed state. The controller **122** is communicable with at least one of the vent valve **118** or the compressed air pump **120**. In various embodiments, the vent valve **118** is an electropneumatic valve.

[0019] The hydrogen fuel system **104** may additionally include an inert gas source (e.g., tank, etc.) that is configured to provide an inert gas (e.g., nitrogen, etc.) to the vent valve **118** to purge the hydrogen gas. This purging may be performed, for example, upon shutdown of the hydrogen internal combustion engine system **100**.

[0020] The hydrogen fuel system **104** includes a hydrogen fuel source **106** (e.g., tank, etc.). The hydrogen fuel source **106** contains hydrogen (e.g., hydrogen fuel, etc.). The hydrogen fuel system **104** also includes a plurality of the hydrogen fuel conduits **108** (e.g., hydrogen fuel lines, etc.). At least one of the hydrogen fuel conduits **108** is fluidly coupled to the hydrogen fuel source **106** and is configured to receive the hydrogen from the hydrogen fuel source **106**.

[0021] The hydrogen fuel system **104** also includes a pressure regulator **111** (e.g., low pressure regulator, etc.). The pressure regulator **111** is fluidly coupled to at least one of the hydrogen fuel conduits **108** and is configured to receive the hydrogen from the hydrogen fuel source **106** via at

least one of the hydrogen fuel conduits **108**. The pressure regulator **111** is configured to maintain a pressure of the hydrogen in the hydrogen fuel conduits **108** below a regulation pressure threshold. This regulation pressure threshold may be associated with maintained prolonged desirable operation of the hydrogen internal combustion engine system **100**.

[0022] The hydrogen fuel system **104** also includes at least one of the hydrogen fuel rails **112**. In some embodiments, the hydrogen fuel system **104** includes multiple of the hydrogen fuel rails **112**. The hydrogen fuel rails **112** extend along portions of the hydrogen internal combustion engine **102** and deliver the hydrogen fuel to the hydrogen internal combustion engine **102**.

[0023] The hydrogen fuel system **104** also includes at least one hydrogen fuel injector **114**. Each of the hydrogen fuel injectors **114** is coupled to the hydrogen internal combustion engine **102** and at least one of the hydrogen fuel rails **112** and receives the hydrogen from the at least one of the hydrogen fuel rails **112**. Each of the hydrogen fuel rails **112** is configured to provide the hydrogen to at least one of the hydrogen fuel injectors **114**.

[0024] The hydrogen internal combustion engine **102** includes a plurality of cylinders **116**. For example, the hydrogen internal combustion engine **102** includes 4, 6, 8, 10, 12, or other numbers of the cylinders **116**. Additionally, the hydrogen internal combustion engine **102** includes a plurality of pistons. Each of the pistons is located within one of the cylinders **116**. Each of the hydrogen fuel injectors **114** is configured to provide the hydrogen fuel into one of the cylinders **116** for combustion. Combustion within the cylinders **116** is harnessed by pistons of the hydrogen internal combustion engine **102** and produces power.

[0025] While not shown in FIG. 1, it is understood that the hydrogen internal combustion engine system **100** includes various other components (e.g., valves, sensors, shafts, pulleys, radiator, etc.) that facilitate combustion of the hydrogen fuel and operation of the hydrogen internal combustion engine **102**.

[0026] The hydrogen fuel system **104** also includes the vent valve **118** (e.g., relief valve, ball valve, manual valve, butterfly valve, pneumatic valve, solenoid valve, etc.). The vent valve **118** is fluidly coupled to one of the hydrogen fuel conduits **108**. As is explained in greater detail herein, and illustrated specifically in FIGS. 2 and 5, the vent valve **118** may be variously located so as to tailor the hydrogen fuel system **104** to a particular application. For example, the vent valve **118** may be fluidly coupled to one of the hydrogen fuel conduits **108** that extends to the hydrogen fuel rail **112**. Alternatively, the vent valve **118** may be fluidly coupled to the hydrogen fuel rail **112** (e.g., proximate an outlet of the hydrogen fuel rail **112**, etc.).

[0027] The vent valve **118** is operable between a closed state (e.g., a non-venting state, a non-purging state, etc.) and an open state (e.g., a venting state, a purging state, etc.). When the vent valve **118** is in the open state, the vent valve **118** vents (e.g., releases, etc.) a portion of the hydrogen within the hydrogen fuel system **104** to atmosphere (e.g., to an environment surrounding the hydrogen internal combustion engine system **100**, etc.). In this way, the vent valve **118** is utilized to vent hydrogen and depressurize the hydrogen fuel system **104**. Venting and depressurizing the hydrogen fuel system **104** may be desirable at shutdown and/or during maintenance of the hydrogen internal combustion engine system **100**. The vent valve **118** is also capable of purging the hydrogen fuel system **104** (e.g., purging the hydrogen from the hydrogen fuel system **104**, etc.) when in the open state.

[0028] By venting the hydrogen using the vent valve **118**, the hydrogen internal combustion engine system **100** may be capable of accommodating long shut-down periods and may also be capable of maintaining operation for prolonged periods of time, such as during contingency operation. Additionally, venting the hydrogen using the vent valve **118** mitigates leakage of the hydrogen from the hydrogen fuel system **104** (e.g., unintended leakage of hydrogen from components of the hydrogen fuel system **104**, etc.), which makes the hydrogen internal combustion engine system **100** more desirable than other systems which do not vent hydrogen. During venting, low pressure hydrogen may be provided to the hydrogen fuel injectors **114** while high pressure hydrogen is

vented. This enables operation of the hydrogen internal combustion engine **102** to be maintained during venting.

[0029] As is discussed in more detail herein, the vent valve **118** may be transitioned between the closed state and the open state in various ways. For example, the vent valve **118** may be manually operated (e.g., by a user, etc.) so as to transition between the closed state and the open state. The vent valve **118** may instead be pneumatically operated (e.g., using pneumatic pressure, using air pressure from the compressed air pump **120**, etc.) so as to transition between the closed state and the open state. In another example, the vent valve **118** may be electronically operated (e.g., by the controller **122**, etc.) so as to transition between the closed state and the open state.

[0030] In various embodiments, the hydrogen fuel system **104** also includes a compressed air pump **120** (e.g., air compressor, etc.). The compressed air pump **120** is fluidly coupled to the vent valve **118** (e.g., via an air conduit.). In these embodiments, compressed air from the compressed air pump **120** is utilized to transition the vent valve between the closed state and the open state. In other embodiments, the hydrogen fuel system **104** does not include the compressed air pump **120**.

[0031] The controller **122** includes a processing circuit **124**. The processing circuit **124** includes a processor **126**. The processing circuit **124** also includes a memory **128**. In various embodiments, the controller **122** is communicable with the vent valve **118**. In these embodiments, the controller **122** is configured to cause the vent valve **118** to transition between the closed state and the open state (e.g., to vent the hydrogen, etc.). In embodiments where the compressed air pump **120** is included, the controller **122** is communicable with the compressed air pump **120** and configured to cause the vent valve **118** to transition between the closed state and the open state (e.g., to vent the hydrogen, etc.) using the compressed air pump **120**. When the vent valve **118** is in the open state, venting of the hydrogen occurs until pressure of the hydrogen is equal to atmospheric pressure or until the vent valve **118** is transitioned from the open state to the closed state.

[0032] FIG. 2 illustrates the hydrogen fuel system **104** according to an embodiment. In this embodiment, the vent valve **118** is fluidly coupled to one of the hydrogen fuel conduits **108** that extends between the hydrogen fuel source **106** and the hydrogen fuel rail **112**. As a result, the vent valve **118** is capable of venting the hydrogen upstream of the hydrogen fuel rail **112**. In various embodiments, the compressed air pump **120** is included and is fluidly coupled to the vent valve **118**. In other embodiments, the compressed air pump **120** is not included.

[0033] FIG. 3 illustrates the vent valve **118** being a manually operated valve. The vent valve **118** includes a lever **300** (e.g., a handle, etc.). The lever **300** is manually operated to open an internal mechanism **302** (e.g., ball, disc, etc.) of the vent valve **118**. For example, the lever **300** may be coupled to the internal mechanism **302** such that manual operation of the lever **300** selectively adjusts the internal mechanism **302** to transition the vent valve **118** between the open state and the closed state. The internal mechanism **302** selectively provides a path for trapped hydrogen in the hydrogen fuel rail **112** and/or the hydrogen fuel conduits **108** to atmosphere. For example, an operator of the hydrogen internal combustion engine system **100** may manually operate the lever **300** to open the internal mechanism **302** to provide the path for the hydrogen within the hydrogen fuel rail **112** and/or the hydrogen fuel conduits **108** to the surroundings of the hydrogen internal combustion engine system **100**.

[0034] FIG. 4 illustrates the vent valve **118** being a pneumatically-actuated valve. The vent valve **118** includes a compressed air inlet **400**. The compressed air inlet **400** is fluidly coupled to the compressed air pump **120** and is configured to receive compressed air from the compressed air pump **120**. The vent valve **118** also includes an internal mechanism **402** (e.g., diaphragm, etc.). The internal mechanism **402** is selectively activated (e.g., lifted, etc.) by the compressed air which causes the vent valve **118** to transition from the closed state to the open state by opening a valve seat and providing a path for trapped hydrogen in the hydrogen fuel rail **112** and/or the hydrogen fuel conduits **108** to atmosphere. For example, the controller **122** may operate the compressed air pump **120** to provide the compressed air to the compressed air inlet **400** to selectively activate the

internal mechanism **402** from the closed state to the open state to provide the path for the hydrogen within the hydrogen fuel rail **112** and/or the hydrogen fuel conduits **108** to the surroundings of the hydrogen internal combustion engine system **100**. In some embodiments, the vent valve **118** is biased towards the closed state such that the vent valve **118** returns to the closed state when the internal mechanism **402** is no longer being selectively activated by the compressed air received from the compressed air pump **120**.

[0035] FIG. **5** illustrates the hydrogen fuel system **104** according to an embodiment. In this embodiment, the vent valve **118** is fluidly coupled to the hydrogen fuel rail **112**. As a result, the vent valve **118** is capable of venting the hydrogen within the hydrogen fuel rail **112**. In various embodiments, the compressed air pump **120** is included and is fluidly coupled to the vent valve **118**. In other embodiments, the compressed air pump **120** is not included.

[0036] FIG. **6** illustrates a method **600** of operating the vent valve **118** by the controller **122**. The method **600** includes, in block **602**, receiving, by the controller **122**, a valve signal. The valve signal may be transmitted to the controller **122** by an engine control unit (ECU) of the hydrogen internal combustion engine system **100**, for example. Additionally or alternatively, the valve signal may be transmitted to the controller **122** by a mobile device (e.g., phone, tablet, laptop, etc.) or network (e.g., a wireless network, etc.).

[0037] The method **600** also includes, in block **604**, transmitting, by the controller **122**, an opening signal (e.g., a first control signal, etc.) corresponding to transitioning the vent valve **118** from the closed state to the open state after receiving the valve signal. For example, the controller **122** may transmit the opening signal responsive to receiving the valve signal. In response to the opening signal, the vent valve **118** is transitioned from the closed state to the open state. For example, the controller **122** may transmit the opening signal to the compressed air pump **120** to cause the compressed air pump **120** to provide compressed air to the vent valve **118** to transition the vent valve **118** from the closed state to the open state. In another example, the controller **122** may transmit the opening signal to the vent valve **118** such that the vent valve **118** itself may be caused to transition from the closed state to the open state (e.g., where the vent valve **118** is an electropneumatic valve, etc.). After the vent valve **118** is opened, the hydrogen is vented from the hydrogen fuel system **104**.

[0038] The method **600** includes, in block **606**, receiving, by the controller **122**, a sealing signal (e.g., a second control signal, etc.) after transmitting the opening signal. The sealing signal may be transmitted to the controller **122** by the ECU of the hydrogen internal combustion engine system **100**, for example. Additionally or alternatively, the sealing signal may be transmitted to the controller **122** by a mobile device (e.g., phone, tablet, laptop, etc.) or network. The sealing signal may be transmitted to the controller **122** upon start-up of the hydrogen internal combustion engine system **100** (e.g., by the ECU, etc.).

[0039] The method **600** also includes, in block **608**, transmitting, by the controller **122**, a closing signal corresponding to transitioning the vent valve **118** from the open state to the closed state after receiving the sealing signal. For example, the controller **122** may transmit the closing signal responsive to receiving the sealing signal. In response to the closing signal, the vent valve **118** is transitioned from the open state to the closed state. For example, the controller **122** may transmit the closing signal to the compressed air pump **120** to cause the compressed air pump **120** to cease providing compressed air to the vent valve **118** to transition the vent valve **118** from the open state to the closed state. In another example, the controller **122** may transmit the closing signal to the vent valve **118** such that the vent valve **118** itself may be caused to transition from the open state to the closed state (e.g., where the vent valve **118** is an electropneumatic valve, etc.). After the vent valve **118** is closed, the hydrogen is no longer vented from the hydrogen fuel system **104**. The method **600** may then return to block **602**.

[0040] While this specification contains many specific implementation details, these should not be construed as limitations on the scope of what may be claimed but rather as descriptions of features

specific to particular implementations. Certain features described in this specification in the context of separate implementations can also be implemented in combination in a single implementation. Conversely, various features described in the context of a single implementation can also be implemented in multiple implementations separately or in any suitable subcombination. Moreover, although features may be described as acting in certain combinations and even initially claimed as such, one or more features from a claimed combination can, in some cases, be excised from the combination, and the claimed combination may be directed to a subcombination or variation of a subcombination.

[0041] As utilized herein, “generally” and similar terms are intended to have a broad meaning in harmony with the common and accepted usage by those of ordinary skill in the art to which the subject matter of this disclosure pertains. It should be understood by those of skill in the art who review this disclosure that these terms are intended to allow a description of certain features described and claimed without restricting the scope of these features to the precise numerical ranges provided. Accordingly, these terms should be interpreted as indicating that insubstantial or inconsequential modifications or alterations of the subject matter described and claimed are considered to be within the scope of the appended claims.

[0042] The term “coupled” and the like, as used herein, mean the joining of two components directly or indirectly to one another. Such joining may be stationary (e.g., permanent) or moveable (e.g., removable or releasable). Such joining may be achieved with the two components or the two components and any additional intermediate components being integrally formed as a single unitary body with one another, with the two components, or with the two components and any additional intermediate components being attached to one another.

[0043] It is important to note that the construction and arrangement of the various systems shown in the various example implementations is illustrative only and not restrictive in character. All changes and modifications that come within the spirit and/or scope of the described implementations are desired to be protected. It should be understood that some features may not be necessary, and implementations lacking the various features may be contemplated as within the scope of the disclosure, the scope being defined by the claims that follow. When the language “a portion” is used, the item can include a portion and/or the entire item unless specifically stated to the contrary.

[0044] Any references herein to the positions of elements are merely used to describe the orientation of various elements in the FIGURES. It should be noted that the orientation of various elements may differ according to other exemplary embodiments, and that such variations are intended to be encompassed by the present disclosure.

[0045] Also, the term “or” is used, in the context of a list of elements, in its inclusive sense (and not in its exclusive sense) so that when used to connect a list of elements, the term “or” means one, some, or all of the elements in the list. Conjunctive language such as the phrase “at least one of X, Y, and Z,” unless specifically stated otherwise, is otherwise understood with the context as used in general to convey that an item, term, etc. may be either X, Y, Z, X and Y, X and Z, Y and Z, or X, Y, and Z (i.e., any combination of X, Y, and Z). Thus, such conjunctive language is not generally intended to imply that certain embodiments require at least one of X, at least one of Y, and at least one of Z to each be present, unless otherwise indicated.

Claims

1. A hydrogen fuel system comprising: a hydrogen fuel conduit; a hydrogen fuel rail in fluid communication with the hydrogen fuel conduit; and a vent valve in fluid communication with at least one of the hydrogen fuel conduit or the hydrogen fuel rail, the vent valve configured to selectively transition between an open state and a closed state, the open state facilitating at least one of venting or purging of the hydrogen fuel system.

2. The hydrogen fuel system of claim 1, wherein the vent valve is in fluid communication with the hydrogen fuel conduit upstream of the hydrogen fuel rail.
3. The hydrogen fuel system of claim 2, further comprising: a pressure regulator in fluid communication with the hydrogen fuel conduit, wherein the vent valve is in fluid communication with the hydrogen fuel conduit downstream of the pressure regulator.
4. The hydrogen fuel system of claim 1, wherein the vent valve is in fluid communication with the hydrogen fuel rail downstream of the hydrogen fuel conduit.
5. A hydrogen internal combustion engine system comprising: the hydrogen fuel system of claim 1; and a hydrogen internal combustion engine; wherein the hydrogen fuel rail is coupled to the hydrogen internal combustion engine.
6. A hydrogen internal combustion engine system comprising: the hydrogen fuel system of claim 1; and a hydrogen fuel source configured to contain hydrogen; wherein the hydrogen fuel conduit is in fluid communication with the hydrogen fuel source such that the hydrogen fuel conduit receives the hydrogen from the hydrogen fuel source.
7. The hydrogen fuel system of claim 1, further comprising a compressed air pump fluidly coupled to the vent valve, the compressed air pump configured to selectively provide compressed air to the vent valve to transition the vent valve between the open state and the closed state.
8. The hydrogen fuel system of claim 7, further comprising a controller communicable with the vent valve, the controller configured to: receive a valve signal; and responsive to receiving the valve signal, transmit, to the compressed air pump, an opening signal such that the vent valve is transitioned from the closed state to the open state.
9. A hydrogen internal combustion engine system comprising: the hydrogen fuel system of claim 8; a hydrogen internal combustion engine coupled to the hydrogen fuel system; and an engine control unit configured to control the hydrogen internal combustion engine, wherein the controller receives the valve signal from the engine control unit.
10. The hydrogen fuel system of claim 8, wherein the controller is configured to: receive a sealing signal; and responsive to receiving the sealing signal, transmit, to the compressed air pump, a closing signal such that the vent valve is transitioned from the open state to the closed state.
11. The hydrogen fuel system of claim 1, further comprising a controller communicable with the vent valve, the controller configured to cause the vent valve to transition between the open state and the closed state, wherein the vent valve is an electropneumatic valve.
12. The hydrogen fuel system of claim 11, wherein the controller is configured to: receive a valve signal; and responsive to receiving the valve signal, transmit, to the vent valve, an opening signal such that the vent valve is transitioned from the closed state to the open state.
13. A hydrogen internal combustion engine system comprising: the hydrogen fuel system of claim 12; a hydrogen internal combustion engine coupled to the hydrogen fuel system; and an engine control unit configured to control the hydrogen internal combustion engine, wherein the controller receives the valve signal from the engine control unit.
14. The hydrogen fuel system of claim 12, wherein the controller is configured to: receive a sealing signal; and responsive to receiving the sealing signal, transmit, to the vent valve, a closing signal such that the vent valve is transitioned from the open state to the closed state.
15. The hydrogen fuel system of claim 1, wherein the vent valve comprises a lever configured to be manually operated by an operator of the hydrogen fuel system to transition the vent valve between the open state and the closed state.
16. The hydrogen fuel system of claim 15, wherein the vent valve comprises an internal mechanism coupled to the lever, the internal mechanism configured to be adjusted by the lever such that the internal mechanism selectively provides a path through the vent valve when the vent valve is in the open state.
17. A method for operating a hydrogen fuel system, the method comprising: receiving a valve signal; and responsive to receiving the valve signal, transmitting, to a component of the hydrogen

fuel system, an opening signal that causes a vent valve of the hydrogen fuel system to transition from a closed state to an open state, the open state facilitating at least one of venting or purging of the hydrogen fuel system.

18. The method of claim 17, further comprising: receiving the valve signal from an engine control unit configured to control a hydrogen combustion engine coupled to the hydrogen fuel system.

19. The method of claim 17, further comprising: receiving a sealing signal; and responsive to receiving the sealing signal, transmitting, to the component, a closing signal that causes the vent valve to transition from the open state to the closed state.

20. The method of claim 19, further comprising: receiving the valve signal from an engine control unit configured to control a hydrogen combustion engine coupled to the hydrogen fuel system; and receiving the sealing signal from the engine control unit upon start-up of the hydrogen combustion engine.
